

WORKING PAPER • PART I OF TWO

Structuring Meaning for Machine Cognition

Semantic models, the cognition spectrum, and shared references



where does the cognition sit?

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The management plane is acquiring participants that reason

For most of its history, network management moved information between programs. The data models carried form; the meaning lived in specifications and in the engineers who read them. The reasoning that closed the gap was human.

Now the systems joining the plane reason for themselves. A machine cognitive consumer needs more than a well-formed document. It needs to comprehend what the document means — the parts a human would have supplied from context.

THEN

Form for programs

Software produces and consumes schemas. A human supplies the meaning at each end.

NOW

Meaning for comprehension

A reasoning consumer must comprehend the model directly, without a human in the loop.

**Two systems must exchange information.
The exchange must be designed.
Design requires cognition.**

So where does the cognition lie?

For decades the answer was uniform: cognition was a human function. Engineers read the schemas, designed the exchange, and settled differences by hand. Once some systems reason for themselves, the answer stops being uniform — and that difference governs how a reconciliation can proceed.

How many of the two systems are interrogable agents?

Both cognitive



Cognition sits in both systems

Confirmation is bilateral

No residual

Self-ratified by both parties

The conversation is the mediation

One inert



Cognition in the one live system

Evidence, no confirmer

Residual: the inert side's opaque parts

Optional human ratification

One system mediates for a silent other

Both inert



Cognition above both, a third party

None; candidates only

Residual on both sides

External ratification, required

A third party proposes; not a party

As cognition recedes, more is inferred without confirmation: confidence falls, the residual grows, and ratification passes outward to a human. The method does not change — only what it costs.

Two takes on the same material

THE PARTS OF MEANING

Pragmatics

how it is used — use, realm, state, authority, adequacy, context

Ontology + lexicon

the entities and relations, and the terms that name and disambiguate them

Provenance — who asserted it, by what method, how firmly

Semantic model = ontology (incl. lexicon) + pragmatics

Grounded in the semiotic triad: syntax (form, set aside), semantics, pragmatics.

THE DIKW LADDER

Wisdom = cognition itself

consumes, manipulates, generates knowledge

Knowledge = the semantic model

information + pragmatics

Information

data under an ontology + lexicon

Data

values, under a schema

The lift carries data up to knowledge; the machine supplies the wisdom.

Completeness for comprehension

A representation is complete when a machine can comprehend and act on it without a human supplying what the form leaves out. That completion is the semantic model. It is self-contained and portable, read as meaning without parsing any form — and ad hoc, not universal: built to purpose, never a standing model of the domain.

PRAGMATICS, RECORDED AS FIRST-CLASS CONTENT

● Use

what a value is for

● Realm

whose decision it reflects

● State

intended vs. operational

● Authority

source of truth on conflict

● Adequacy

how much fidelity is enough

● Context

conditions that change meaning

● Provenance

source, method, confidence

Completing a representation — before any second system

The lift produces a semantic model from a data model. It is worth naming on its own: it is what makes a single model comprehensible to a machine, whether or not anything is ever reconciled against it. Much of the value is here, in completion alone.

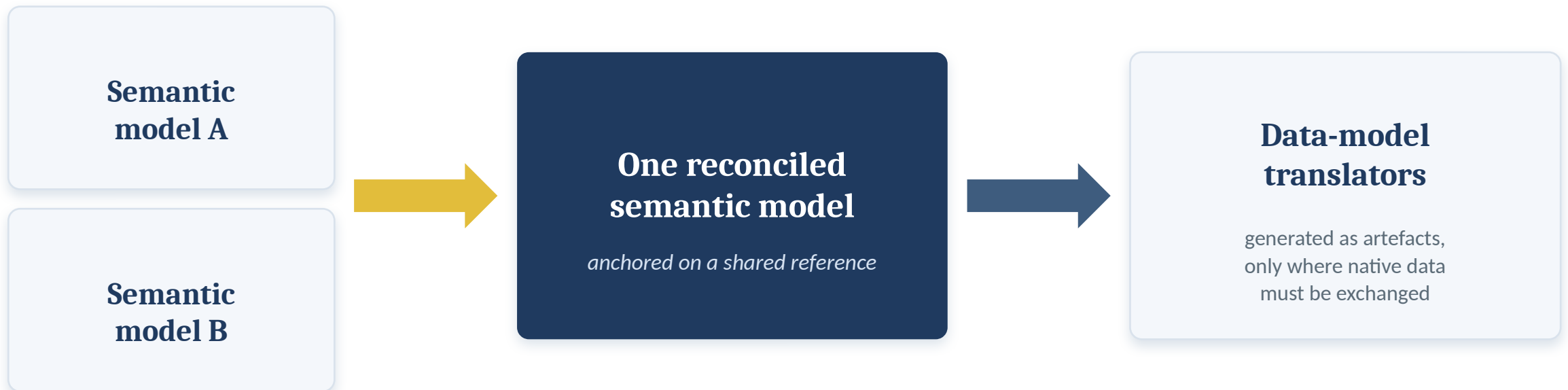


Where the system can be interrogated, the lift is helped by the system explaining itself. Where it cannot, it proceeds from schema and instances alone, holds results with stated confidence, and leaves a residual — exactly the cognition spectrum.

RECONCILIATION

Making two semantic models correspond

Reconciliation is carried out at the level of the semantic models — aligning ontologies, lexicons, and pragmatics, where the meaning lives. The product is one reconciled model that merges, unites, or bridges the two.



Bind once per system: N bindings, not N^2 translators.

Named by the part of the model — and checked to fit

DIVERGENCES FALL IN ONE OR MORE OF THREE PARTS

Lexicon

naming — same concept, different labels, or a false cognate

Ontology

how the world is carved — granularity, scope, abstraction

Pragmatics

how it is used — intended use, authority, context

Every reconciliation is in this sense cross-domain. An abstraction gap usually shows up twice — ontological and pragmatic at once.

VERIFICATION STRENGTHENS TOWARD THE COGNITIVE END

● Both cognitive

closed-loop worked exchange, self-ratified

● One inert

evidence + round-trip on invariants

● Both inert

external adjudication or downstream test

Thin, practical anchors that assist cognition

A thin lexical reference is an identity-only anchor: an id, a preferred label and synonyms, a shallow class, a disambiguating definition, one canonical example. Correspondences arrive as near-identity; false cognates are pre-empted; and its value grows as machine cognition recedes.

A FAMILY OF REFERENCES, BY THE FACET EACH PINS

Thin lexical

what a term means

Instance / identity

which individual

Unit / measurement

how much, in what terms

Value-set / enumeration

discrete values & ordering

Structural

shape

Pragmatic

use and context

Provenance / trust

confidence, source, method

Canonical scenario

meaning by example

Invariant

what must be preserved

Four cases chosen to span an operational space

● Configuration

Two peer models of one transport network. A false cognate — link- vs tunnel-termination — pre-empted by a thin reference. Verified by round-trip on invariants.

configuration.html

● Intent

A declarative quantitative intent refined to a realisation. Not equivalence but satisfaction; conflicts resolved by a best-achievable offer.

intent.html

● Cross-domain

Two bespoke systems, no shared standard. A minimal reference built between them; every requirement pinned to a canonical example.

cross_domain.html

● Observability

Legacy fault model vs. explicit anomaly semantics. Alarm $\neq$ anomaly; meaning carried largely by pragmatics.

anomaly.html

The study's objectives

- 1 **Find** the shared references that are most effective, practical, and flexible
- 2 **Quantify** their benefit across realistic cases and across the cognition spectrum
- 3 **Delineate** the cognitive workflows of reconciliation, and measure their intensity
- 4 **Locate** where a reference reduces the requirement for human supervision

Part II · a research programme to test and quantify these claims